LAPORAN PRAKTIKUM
WORKSHOP SISTEM INFORMASI WEB SERVER
(TIF330805)
SEMESTER III



Acara 17
API CRUD

NAMA :

ANGELLO ERIYA DEWANGGA (E41250776)

PROGRAM STUDI DILUAR KAMPUS UTAMA (PSDKU)

TEKNIK INFORMATIKA KAB. NGANJUK

JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI JEMBER

TAHUN 2026

A. Tujuan Pengujian

Pengujian dilakukan untuk memastikan endpoint API mahasiswa dapat menjalankan operasi GET, POST, PUT, dan DELETE dengan benar serta menerapkan autentikasi JSON Web Token (JWT) pada endpoint yang memerlukan akses terproteksi.

Pengujian dilakukan menggunakan **curl.exe** melalui terminal PowerShell sebagai alternatif dari Postman.

1. Metode GET

```
$base = "http://localhost/BKPM/Acara17/si-
akademik/api/mahasiswa"
curl.exe -i $base
```

2. Metode POST

```
$body = '{"nim":"2024004","nama":"Budi
Uji","email":"budiuji@mail.com","jurusan":"Teknik
Informatika"}'
[System.IO.File]::WriteAllText(
    "$PWD\body.json",
    $body,
    [System.Text.UTF8Encoding]::new($false)
)
curl.exe -i -X POST $base `
  -H "Content-Type: application/json" `
  --data-binary "@body.json"
```

3. Metode PUT

```
$body = '{"nim":"2024001","nama":"Budi Santoso
Updated","email":"budi@mail.com","jurusan":"Sistem
Informasi"}'
[System.IO.File]::WriteAllText(
```

```
    "$PWD\body.json",
    $body,
    [System.Text.UTF8Encoding]::new($false)
)
curl.exe -i -X PUT "${base}?id=1" `
  -H "Content-Type: application/json" `
  --data-binary "@body.json"
```

4. Metode DELETE

```
curl.exe -i -X DELETE "${base}?id=1"
```

- **$base** digunakan untuk menyimpan URL endpoint API mahasiswa agar dapat digunakan kembali pada perintah berikutnya.
- **http://localhost** menunjukkan bahwa aplikasi dijalankan pada komputer lokal.
- **/BKPM/Acara17/si-akademik/api/mahasiswa** merupakan alamat endpoint yang menangani data mahasiswa
- **curl.exe** digunakan untuk mengirim permintaan HTTP ke server.
- **-X POST** menentukan bahwa permintaan HTTP menggunakan metode POST untuk mengirim data baru ke API.
- **-X PUT** menentukan bahwa permintaan HTTP digunakan untuk memperbarui data.
- **-X DELETE** menentukan bahwa permintaan HTTP digunakan untuk menghapus data.
- **-i** digunakan untuk menampilkan header respons HTTP beserta isi respons dari server.
- **-H "Content-Type: application/json**" memberi tahu server bahwa isi permintaan menggunakan format JSON.
- **--data-binary "@body.json"** mengirim isi file body.json sebagai badan permintaan (request body).

B. Hasil Pengujian

```
PS C:\laragon\www\BKPM\Acara17> $base = "http://localhost/BKPM/Acara17/si-akademik/public/mahasiswa"
>>
>> $body = '{"nim":"2201001","nama":"Andi Saputra","prodi":"Teknik Informatika"}'
>>
>> [System.IO.File]::WriteAllText(
>>     "$PWD\body.json",
>>     $body,
>>     [System.Text.UTF8Encoding]::new($false)
>> )
PS C:\laragon\www\BKPM\Acara17> curl.exe -i -X POST "$base" `
>>   -H "Content-Type: application/json" `
>>   --data-binary "@body.json"
HTTP/1.1 201 Created
Date: Tue, 20 Oct 2026 17:18:40 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Content-Length: 72
Content-Type: application/json

{"success":true,"message":"Data berhasil ditambahkan","data":{"id":"3"}}
PS C:\laragon\www\BKPM\Acara17>
```

1. Membuat request baru dengan method POST

```
PS C:\laragon\www\BKPM\Acara17> curl.exe -s "$base" | ConvertFrom-Json | ConvertTo-Json -Depth 10
{
    "success":  true,
    "data":  [
                 {
                     "id":  3,
                     "nim":  "2201001",
                     "nama":  "Andi Saputra",
                     "prodi":  "Teknik Informatika",
                     "created_at":  "2026-10-21 00:18:40"
                 },
                 {
                     "id":  2,
                     "nim":  "2024002",
                     "nama":  "Siti Aminah",
                     "prodi":  "Sistem Informasi",
                     "created_at":  "2026-10-20 23:15:14"
                 },
                 {
                     "id":  1,
                     "nim":  "2024001",
                     "nama":  "Budi Santoso",
                     "prodi":  "Teknik Informatika",
                     "created_at":  "2026-10-20 23:15:14"
                 }
             ]
}
PS C:\laragon\www\BKPM\Acara17>
```

2. Menguji Method GET

```
PS C:\laragon\www\BKPM\Acara17> $base = "http://localhost/BKPM/Acara17/si-akademik/public/mahasiswa"
PS C:\laragon\www\BKPM\Acara17> $body = '{"nim":"2024001","nama":"Budi Santoso Updated","prodi":"Sistem Informasi"}'
>>
>> [System.IO.File]::WriteAllText(
>>     "$PWD\body.json",
>>     $body,
>>     [System.Text.UTF8Encoding]::new($false)
>> )
>>
>> curl.exe -i -X PUT "${base}?id=1" `
>>   -H "Content-Type: application/json" `
>>   --data-binary "@body.json"
HTTP/1.1 200 OK
Date: Wed, 21 Oct 2026 01:53:38 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Content-Length: 53
Content-Type: application/json

{"success":true,"message":"Data berhasil diperbarui"}
```

3. Menguji Method PUT menggunakan data baru

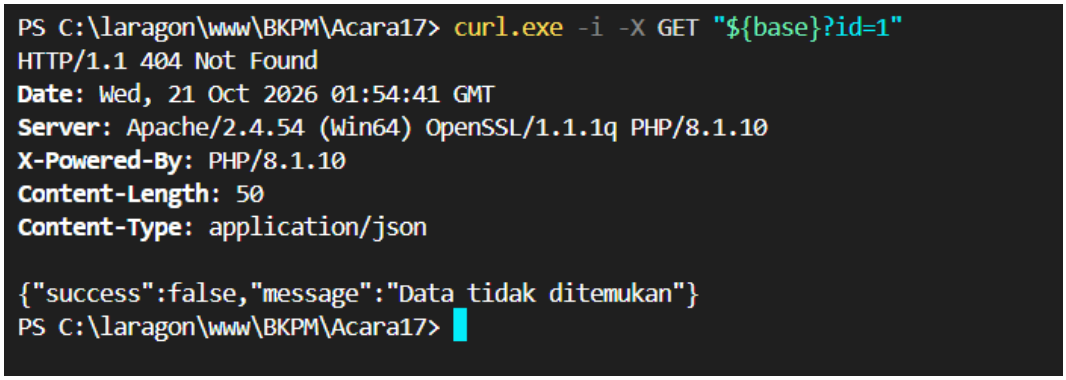
```
PS C:\laragon\www\BKPM\Acara17> curl.exe -s "${base}?id=1" |
>>   ConvertFrom-Json |
>>   ConvertTo-Json -Depth 10
{
    "success":  true,
    "data":  {
                 "id":  1,
                 "nim":  "2024001",
                 "nama":  "Budi Santoso Updated",
                 "prodi":  "Sistem Informasi",
                 "created_at":  "2026-10-20 23:15:14"
             }
}
PS C:\laragon\www\BKPM\Acara17>
```

4. Hasil dari method PUT ke ID 1

```
PS C:\laragon\www\BKPM\Acara17> curl.exe -i -X DELETE "${base}?id=1"
HTTP/1.1 200 OK
Date: Wed, 21 Oct 2026 01:54:25 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Content-Length: 50
Content-Type: application/json

{"success":true,"message":"Data berhasil dihapus"}
PS C:\laragon\www\BKPM\Acara17>
```

5. Menguji method DELETE



6. Hasil dari method DELETE, data yang dihapus sudah tidak ditemukan